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Green's Theorem.

Let R be a simply connected plane region whose boundary is a simple closed piecewise smooth curve C oriented counterclockwise. If $f(x, y)$ and $g(x, y)$ are continuous and have continuous first partial derivatives on some open set containing R , then

$$\oint_C f(x, y) dx + g(x, y) dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

1) Using Green's theorem evaluate

$$\int_C (xy + y^2) dx + x^2 dy \text{ where } C \text{ is the}$$

boundary of the area common to the curve $y = x^2$ & $y = x$

$$f = xy + y^2$$

$$g = x^2$$

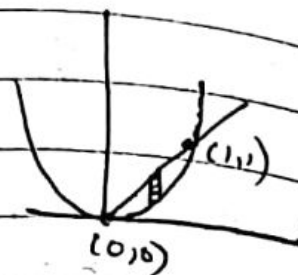
$$\frac{\partial f}{\partial y} = x + 2y$$

$$\frac{\partial g}{\partial x} = 2x$$

$$\int_C (xy + y^2) dx + x^2 dy = \iint 2x - (x + 2y) dA$$

3)

$$= \int_0^1 \int_{x^2}^x (x - 2y) dy dx$$



$$= \underline{\underline{-\frac{1}{20}}}$$

$$x: 0 \text{ to } 1$$

$$y: x^2 \rightarrow x$$

4Q)

2) Use Green's Theorem to evaluate

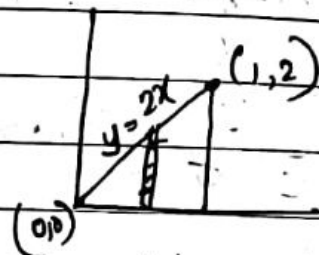
Ans:-
Green's
Theorem

$\int_C x^2 y dx + x dy$ along the Aular path

shown in figure

Ans $\frac{\partial f}{\partial y} = x^2$ $\frac{\partial g}{\partial x} = 1$

$$\int_C x^2 y dx + x dy = \iint (1 - x^2) dx$$



$$= \int_0^1 \int_0^{2x} (1 - x^2) dy dx$$

$$= \underline{\underline{\frac{1}{2}}}$$

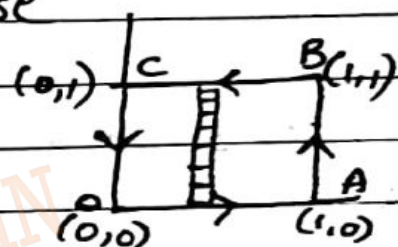
Evaluate the line integral using Green's theorem and check the answer by evaluating directly

$\oint y^2 dx + x^2 dy$ where C is the square with vertices $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$ oriented counterclockwise

$$\oint y^2 dx + x^2 dy$$

$$= \iint 2x - 2y \, dA$$

$$= \int_0^1 \int_0^1 2x - 2y \, dy \, dx = \underline{\underline{0}}$$



$$\text{Also } \oint y^2 dx + x^2 dy = \int_{OA} + \int_{AB} + \int_{BC} + \int_{CO}$$

Along OA $y=0, dy=0$

$$\int_{OA} y^2 dx + x^2 dy = 0$$

Along AB $x=1, dx=0$ $\int_{AB} y^2 \cdot 0 + 1 dy = \int_0^1 dy = \underline{\underline{1}}$

Along BC $y=1, dy=0$

$$\int_{BC} y^2 dx + x^2 dy = \int_1^0 dx = -1$$

Along CO $x=0, dx=0$

$$\int_{CO} y^2 \cdot 0 + 0 \cdot dy = 0$$

$$\therefore \int_C y^2 dx + x^2 dy = 0 + 1 - 1 + 0 = \underline{\underline{0}}$$

Q. 5.

11Q)

4) Apply Green's theorem to evaluate

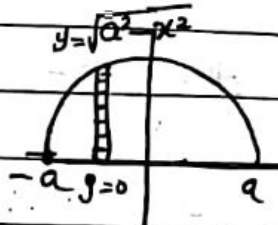
$\int_C (2x^2 - y^2) dx + (x^2 + y^2) dy$ where C is the boundary of the area enclosed by x axis and the upper half of the circle $x^2 + y^2 = a^2$

Ans

Ans

$$\frac{\partial f}{\partial y} = -2y$$

$$\frac{\partial g}{\partial x} = 2x$$



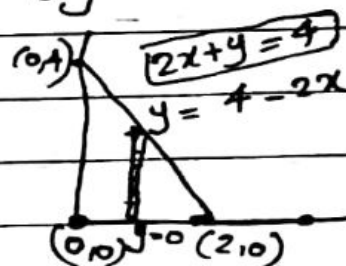
$$\oint_C = \iint_{-a}^a \int_0^{\sqrt{a^2-x^2}} (2x + 2y) dy dx$$

$$\begin{aligned}
 &= \int_{-a}^a \left(2xy + \frac{2y^2}{2} \right) \sqrt{a^2 - x^2} dx \\
 &= \int_{-a}^a \underbrace{2x\sqrt{a^2 - x^2}}_{\text{odd}} + \underbrace{(a^2 - x^2)}_{\text{even}} dx \\
 &= 2 \int_0^a (a^2 - x^2) dx = 2 \left(a^2 x - \frac{x^3}{3} \right)_0^a \\
 &= 2 \left(a^3 - \frac{a^3}{3} \right) = \underline{\underline{\frac{4a^3}{3}}}
 \end{aligned}$$

$\oint_C \ln(1+y) dx - \frac{xy}{1+y} dy$ where C is the triangle with vertices $(0,0)$, $(2,0)$, $(0,4)$

$$\oint_C \ln(1+y) dx - \frac{xy}{1+y} dy = \iint \frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} dA$$

$$\begin{aligned}
 f &= \ln(1+y) & g &= -\frac{xy}{1+y} \\
 \frac{\partial f}{\partial y} &= \frac{1}{1+y} & \frac{\partial g}{\partial x} &= -\frac{y}{1+y}
 \end{aligned}$$



$$\begin{aligned}
 \therefore \oint_C \ln(1+y) dx - \frac{xy}{1+y} dy &= \iint \frac{-y-1}{1+y} dA \\
 &= \int_0^2 \int_0^{4-2x} -1 dy dx = \int_0^2 -(4-2x) dx \\
 &= \underline{\underline{-4}}
 \end{aligned}$$

Q6. $\oint (e^x + y^2)dx + (e^y + x^2)dy$ where C is the bdy of region b/w $y=x^2$, $y=2x$

Ans = $-8/5$

7. $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the bdy of region defined by $x=0$, $y=0$, $x+y=1$ Ans = $5/3$

8. $\oint_C 4xydx + 2xydy$ C is the rectangle bdd by $x=-2$, $x=4$, $y=1$, $y=2$ counterclockwise

Ans = -6

9. $\oint x \cos y dx - y \sin x dy$ where C is the square with vertices $(0,0)$, $(\pi,0)$, (π,π) , $(0,\pi)$.
Ans: $-\pi^2$

10. $\oint (x^2 - 3y)dx + 3x dy$ C : Circle $x^2 + y^2 = 4$
Ans: 24π

11. $\oint \tan^{-1} y dx - \frac{y^2 x}{1+y^2} dy$ C square $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$ By $\oint$
Ans: -1

Note :-

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Parametric eqn of ellipse $x = a \cos t, y = b \sin t$

parametric eqn of circle $x = a \cos t, y = a \sin t$
 $x^2 + y^2 = a^2$ $0 \leq t \leq 2\pi$

Area Using Green's theorem.

$$A = \frac{1}{2} \oint_C -y dx + x dy$$

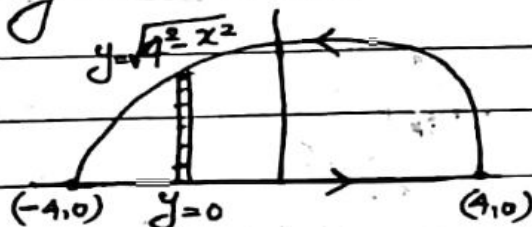
$$\oint_C -y dx + x dy = \iint_D 2 dx dy = 2A$$

Use Green's theorem to find work done by the force field $F(x, y) = xy \mathbf{i} + \left(\frac{x^2}{2} + xy\right) \mathbf{j}$ the particle starts at $(-4, 0)$ traverses the upper semicircle $x^2 + y^2 = 16$ & returns to the starting point along x axis

$$W = \oint_C F \cdot d\mathbf{r}$$

$$= \oint_C xy dx + \left(\frac{x^2}{2} + xy\right) dy$$

green's theorem $= \int_{-4}^4 \int_0^{\sqrt{16-x^2}} x + y - x dy dx = \int_{-4}^4 \frac{y^2}{2} dx = \int_{-4}^4 \frac{(4^2 - x^2)}{2} dx$



$$A = \frac{1}{2} \oint_C -y dx + x dy$$

~~proof~~ By Green's theorem

$$\oint_C -y dx + x dy = \iint \frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} dA$$

$$= \iint 1 - (-1) dA$$

$$= \iint 2 dA$$

$$= 2 \iint dA$$

$$\therefore \iint dA = \frac{1}{2} \oint_C -y dx + x dy$$

Area = " Hence proved.

$$f = -y$$

$$g = x$$

$$\frac{\partial f}{\partial y} = -1$$

$$\frac{\partial g}{\partial x} = 1$$

$$= \frac{2}{2} \left[16x - \frac{x^3}{3} \right]_0^4$$

$$= 64 - \frac{64}{3} = \frac{64 \times 2}{3} = \underline{\underline{\frac{128}{3}}}$$

2. Use line integral to find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Ans: Area $= \frac{1}{2} \oint -y dx + x dy$

$$= \frac{1}{2} \int_0^{2\pi} [-b \sin t (-a \sin t) dt + a \cos t (b \cos t) dt]$$

$$= \frac{1}{2} \int_0^{2\pi} ab dt$$

$$= \frac{ab}{2} (t)_0^{2\pi} = \frac{ab \cdot 2\pi}{2} = \underline{\underline{\pi ab}}$$

For an ellipse

$$x = a \cos t$$

$$y = b \sin t$$

$$t: 0 \leq t \leq 2\pi$$

$$dx = -a \sin t dt$$

$$dy = b \cos t dt$$

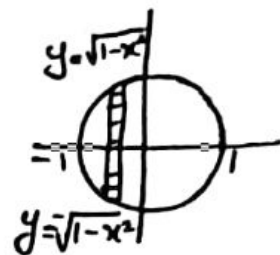
3) Using Green's theorem, find the work done by the force field $F(x, y) = (e^x - y^3)\mathbf{i} + (\cos y + x^3)\mathbf{j}$ on a particle that travels once around the unit circle $x^2 + y^2 = 1$ in the counter clockwise direction

Ans:-

4. Use line integral to find the area of the region enclosed by the astroid

Ans:-

$$\begin{aligned}
 W &= F \cdot dx \\
 &= \oint (e^x y^3) dx + (\cos y - x^3) dy \\
 &= \iint (-3x^2 - 3y^2) dA \\
 &= \iint -3(x^2 + y^2) dy dx \\
 &= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} 3(x^2 + y^2) dy dx \\
 &= 3 \int_{-1}^1 2 \int_0^{\sqrt{1-x^2}} (x^2 y + \frac{y^3}{3}) dy dx \\
 &= 6 \int_{-1}^1 x^2 \sqrt{1-x^2} + \frac{(\sqrt{1-x^2})^3}{3} dx \\
 &= 6 \int_{-1}^1 \sin^2 \theta \cos \theta + \frac{1}{3} (1-x^2)^{3/2} dx \\
 &= 12 \int_0^{\pi/2} \sin^2 \theta \cos \theta + \frac{(1-x^2)^{3/2}}{3} dx \\
 &= 12 \int_0^{\pi/2} (\sin^2 \theta \cos^2 \theta + \frac{\cos^3 \theta}{3}) \cos \theta d\theta \\
 &= 12 \int_0^{\pi/2} (\sin^2 \theta \cos^2 \theta + \frac{\cos^4 \theta}{3}) d\theta = \left[\frac{\sin^3 \theta \cos^2 \theta}{4 \cdot 2} + \frac{3 \cdot 1 \cdot 4}{4 \cdot 2} \right] \frac{\pi}{2} \\
 &= \frac{3 \cdot \pi}{2}
 \end{aligned}$$



$$\begin{aligned}
 x &= \sin \theta \\
 dx &= \cos \theta d\theta \\
 x=0, \theta &= 0 \\
 x=1, \theta &= \pi/2
 \end{aligned}$$

$$\begin{aligned}
 &= \left[\frac{\sin^3 \theta \cos^2 \theta}{4 \cdot 2} + \frac{3 \cdot 1 \cdot 4}{4 \cdot 2} \right] \frac{\pi}{2} \\
 &= \frac{3 \cdot \pi}{2}
 \end{aligned}$$

Surface integral

Any integral which is to be evaluated over a surface is called surface integral

Evaluation of surface integrals.

I Let the surface σ is given by $z = g(x, y)$ then the surface integral is given by

$$\iint_{\sigma} f(x, y, z) ds = \iint_R f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA$$

$$\text{II } \iint_{\sigma} f(x, y, z) ds = \iint_R f(x, g(x, z), z) \sqrt{\left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2 + 1} dA$$

when σ is given by $y = g(x, z)$

$$\text{III } \iint_{\sigma} f(x, y, z) ds = \iint_R f(g(y, z), y, z) \sqrt{\left(\frac{\partial x}{\partial y}\right)^2 + \left(\frac{\partial x}{\partial z}\right)^2 + 1} dA$$

when σ is given by $x = g(y, z)$

When surface given in parametric form

$$\text{ie } \vec{r}(u,v) = x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k}$$

$$\iint_{\sigma} f(x,y,z) ds = \iint_R f(x(u,v), y(u,v), z(u,v)) \left\| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right\| du dv$$

1) Evaluate $\iint f(x,y,z) ds$ over the surface σ represented by the vector field valued function $\vec{r}(u,v)$

$$f(x,y,z) = xyz$$

$$\vec{r}(u,v) = u \cos v \hat{i} + u \sin v \hat{j} + 3u \hat{k} \quad (1 \leq u \leq 3, 0 \leq v \leq \pi/2)$$

$$\text{Q:- } \iint_{\sigma} f(x,y,z) ds = \iint_R f(x(u,v), y(u,v), z(u,v)) \left\| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right\| du dv$$

$$f(x,y,z) = xyz$$

$$= u \cos v \cdot u \sin v \cdot 3u$$

$$= 3u^3 \sin v \cos v$$

$$\frac{\partial \vec{r}}{\partial u} = \cos v \hat{i} + \sin v \hat{j} + 3\hat{k}$$

$$\frac{\partial \gamma}{\partial v} = -u \sin v \hat{i} + u \cos v \hat{j} + 0 \hat{k}$$

$$\frac{\partial \gamma}{\partial u} \times \frac{\partial \gamma}{\partial v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos v & \sin v & 3 \\ -u \sin v & u \cos v & 0 \end{vmatrix}$$

$$= -3u \cos v \hat{i} - 3u \sin v \hat{j} + u \hat{k}$$

$$\left\| \frac{\partial \gamma}{\partial u} \times \frac{\partial \gamma}{\partial v} \right\| = \sqrt{9u^2 \cos^2 v + 9u^2 \sin^2 v + u^2}$$

$$= \sqrt{9u^2 + u^2} = \sqrt{10u^2} = \sqrt{10} u$$

$$\iint_D f(x, y, z) ds = \int_1^3 \int_0^{\pi/2} u \cos v \cdot u \sin v \cdot 3u \cdot \sqrt{10} u \, dv \, du$$

$$= \int_1^3 \int_0^{\pi/2} u^4 \sin v \cos v \cdot 3\sqrt{10} \, dv \, du$$

$$= 3\sqrt{10} \int_1^3 u^4 \left(\frac{\cos 2v}{2} \right) \Big|_0^{\pi/2} dv = \frac{3\sqrt{10}}{4} \int_1^3 u^4 (-2) du$$

$$= \frac{3}{2} \sqrt{10} \left(\frac{u^5}{5} \right) \Big|_1^3 = \frac{3\sqrt{10}}{10} (3^5 - 1^5)$$

$$\approx \frac{363 \sqrt{10}}{5}$$

$$2. f(x, y, z) = \frac{1}{\sqrt{1+4x^2+4y^2}}$$

$$r(u, v) = u \cos v \hat{i} + u \sin v \hat{j} + u^2 \hat{k}$$

$$0 \leq u \leq \sin v, \quad 0 \leq v \leq \pi$$

Ans. $\frac{\partial r}{\partial u} = \cos v \hat{i} + \sin v \hat{j} + 2u \hat{k}$

$$\frac{\partial r}{\partial v} = -u \sin v \hat{i} + u \cos v \hat{j}$$

$$\frac{\partial r}{\partial u} \times \frac{\partial r}{\partial v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos v & \sin v & 2u \\ -u \sin v & u \cos v & 0 \end{vmatrix}$$

$$= \hat{i}(-2u^2 \cos v) - \hat{j}(2u^2 \sin v) + \hat{k}u$$

$$\left\| \frac{\partial r}{\partial u} \times \frac{\partial r}{\partial v} \right\| = \sqrt{4u^4 \cos^2 v + 4u^4 \sin^2 v + u^2}$$

$$= \sqrt{4u^4 + u^2} = u \sqrt{4u^2 + 1}$$

$$\iint_{\sigma} f(x, y, z) d\sigma = \int_0^{2\pi} \int_0^{\sin v} \frac{1}{u \sqrt{4u^2 + 1}} \cdot u \sqrt{4u^2 + 1} du dv$$

$$= \int_0^{2\pi} \int_0^{\sin v} \frac{u}{\sqrt{1+4u^2}} du dv$$

$$= \int_0^{2\pi} \left(\frac{u^2}{2} \right)_0^{\sin v} dv$$

$$= \frac{1}{2} \int_0^{2\pi} \sin^2 v dv$$

$$= \frac{1}{2} \int_0^{2\pi} \frac{1 - \cos 2v}{2} dv = \frac{1}{4} \left(v - \frac{\sin 2v}{2} \right)_0^{2\pi}$$

$$= \frac{1}{4} 2\pi = \underline{\underline{\frac{\pi}{2}}}$$

3. $f(x, y, z) = \frac{x^2 + z^2}{y}$

$$\boxed{8\pi \ln 3}$$

$$r(u, v) = 2\cos v \hat{i} + u \hat{j} + 2\sin v \hat{k} \quad \begin{matrix} 1 \leq u \leq 3 \\ 0 \leq v \leq \pi \end{matrix}$$

4. $f(x, y, z) = e^{-z}$

$$r(u, v) = 2\sin u \cos v \hat{i} + 2\sin u \sin v \hat{j} + 2\cos u \hat{k}$$

$$0 \leq u \leq \pi/2$$

$$0 \leq v \leq \pi$$

$$\boxed{2\pi(1 - e^{-2})}$$

5. $f(x, y, z) = x - y - z$, is the portion of the plane $x + y = 1$ in the 1st octant b/w $z = 0$ & $z = 2$

Ans: $\iint_R f(x, y, z) ds = \iint_R x - y - z \sqrt{\left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2 + 1} dA$

$y = 1 - x$

$\frac{\partial y}{\partial x} = -1, \frac{\partial y}{\partial z} = 0$

$= \int_0^2 \int_0^1 (x - (1 - x) - z) \sqrt{1 + 1} dx dz$

$= \sqrt{2} \int_0^2 \int_0^1 (2x - z - 1) dx dz$

$= \sqrt{2} \int_0^2 \left(\frac{2x^2}{2} - zx - x \right)_0^1 dz$

$= \sqrt{2} \int_0^2 (1 - z - 1) dz$

$= \sqrt{2} \left(-\frac{z^2}{2} \right)_0^2 = -\frac{\sqrt{2}}{2} (4)$

$= -2\sqrt{2}$

6. Evaluate the surface integral $\iint xz \, ds$ where σ is the part of the plane $x+y+z=1$ that lies in the 1st octant

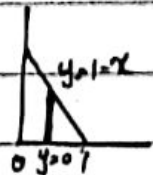
Ans:-

Let $z=1-x-y$.

$$\iint xz \, ds = \iint x(1-x-y) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \, dA$$

$$= \iint x-x^2-xy \sqrt{(-1)^2 + (-1)^2 + 1} \, dA$$

$$= \iint x-x^2-xy \sqrt{3} \, dA$$



$$= \sqrt{3} \int_0^1 \int_0^{1-x} x-x^2-xy \, dy \, dx$$

$$= \sqrt{3} \int_0^1 \left(xy - x^2 y - \frac{xy^2}{2} \right) \Big|_{y=0}^{y=1-x} dx$$

$$= \sqrt{3} \int_0^1 x(1-x) - x^2(1-x) - \frac{x(1-x)^2}{2} dx$$

$$= \sqrt{3} \int_0^1 x-x^2-x^2+x^3-\frac{x}{2}+\frac{x^2}{2}-\frac{x^3}{2} dx$$

$$= \sqrt{3} \int_0^1 \frac{1}{2}x^3 - x^2 + \frac{x}{2} dx = \frac{\sqrt{3}}{24} //$$

7. $\iint x+y \, ds$ where σ is the portion of the plane $z=6-2x-4y$ in the 1st octant

$$\iint_{\sigma} x+y \, ds = \iint_R x+y \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \, dA$$

$$= \iint_R (x+y) \sqrt{(-2)^2 + (-4)^2 + 1} \, dA$$

$$= \iint_R x+y \sqrt{21} \, dA$$

$$= \sqrt{21} \int_0^3 \int_0^{\frac{3-x}{2}} (x+y) \, dy \, dx$$



$$2x+4y=6$$

$$= \sqrt{21} \int_0^3 \left(xy + \frac{y^2}{2} \right) \Big|_0^{\frac{3-x}{2}} dx \quad \begin{matrix} x+y=3 \\ \text{put } z=0 \end{matrix}$$

$$= \sqrt{21} \int_0^3 \left(x \frac{(3-x)}{2} + \frac{1}{2} \left(\frac{(3-x)}{2} \right)^2 \right) dx$$

$$= \frac{\sqrt{21}}{2} \int_0^3 \left(3x - x^2 + \frac{1}{4} (9 - 6x + x^2) \right) dx$$

$$= \frac{\sqrt{21}}{2} \int_0^3 \left(3x - \frac{3x}{2} - x^2 + \frac{x^2}{4} + \frac{9}{4} \right) dx$$

$$= \frac{\sqrt{21}}{2} \int_0^3 \left(\frac{3x^2}{2} + \frac{9}{4} - \frac{3x^2}{4} \right) dx$$

$$= \frac{\sqrt{21}}{2} \left(\frac{3x^3}{2} + \frac{9x}{4} - \frac{3 \cdot \frac{27}{3}}{4} \right)$$

$$= \frac{\sqrt{21}}{2} \left(\frac{3 \cdot 27}{4} + \frac{27}{4} - \frac{27}{4} \right)$$

$$= \frac{27\sqrt{21}}{8}$$

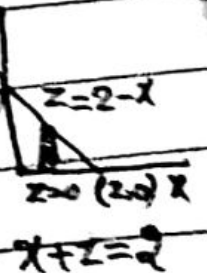
3) $f(x, y, z) = xy$, - is the portion of the plane $x+y+z=2$ lying in the 1st octant

Ans- take $y = 2 - x - z$

$$\iint xy \, ds = \iint x(2-x-z) \sqrt{\left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2 + 1} \, dA$$

$$= \iint 2x - x^2 - xz \sqrt{(-1)^2 + (-1)^2 + 1} \, dA \quad y=0$$

$$= \sqrt{3} \int_0^2 \int_0^{2-x} 2x - x^2 - xz \, dz \, dx$$



$$= \sqrt{3} \int_0^2 \left((2x-x^2)x - x \frac{x^2}{2} \right) dx$$

$$= \sqrt{3} \int_0^2 \left[(2x-x^2)(2-x) - \frac{x(2-x)^2}{2} \right] dx$$

$$= \sqrt{3} \int_0^2 \left[x(2-x)^2 - \frac{x(2-x)^2}{2} \right] dx$$

$$= \sqrt{3} \int_0^2 \frac{1}{2} x(2-x)^2 dx$$

$$= \frac{\sqrt{3}}{2} \int_0^2 x(4-4x+x^2) dx$$

$$= \frac{\sqrt{3}}{2} \int_0^2 (4x-4x^2+x^3) dx$$

$$= \frac{\sqrt{3}}{2} \left(\frac{4x^2}{2} - \frac{4x^3}{3} + \frac{x^4}{4} \right)_0^2$$

$$= \frac{\sqrt{3}}{2} \left(2 \cdot 2^2 - \frac{4 \cdot 2^3}{3} + \frac{2^4}{4} \right)$$

$$= \frac{\sqrt{3}}{2} \left(16 - \frac{32}{3} \right) = \frac{\sqrt{3}}{2} \left(\frac{36-32}{3} \right)$$

$$= \frac{\sqrt{3}}{2} \times \frac{4}{3}$$

$$= \frac{2}{\sqrt{3}}$$

9) $\iint x^2 y \, ds$ where $\rightarrow$ is the portion of the cylinder $x^2 + z^2 = 1$ b/w the planes $y=0$ & $y=1$ and above the xy plane ($z=0$)

$$z^2 = 1 - x^2$$

$$\iint x^2 y \, ds = \iint x^2 y \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}$$

$$= \iint x^2 y \sqrt{\left(\frac{zx}{2\sqrt{1-x^2}}\right)^2 + 0 + 1} \, dA$$

$$= \iint x^2 y \sqrt{\frac{x^2}{1-x^2} + 1} \, dA \quad \begin{matrix} z=0 \\ x^2=1 \end{matrix}$$

$$= \iint x^2 y \sqrt{\frac{x^2 + 1 - x^2}{1-x^2}} \, dA \quad x=\pm 1$$

$$= \int_0^1 \int_{-1}^1 \frac{x^2 y}{\sqrt{1-x^2}} \, dx \, dy \quad \begin{matrix} x = \sin \theta \\ dx = \cos \theta \, d\theta \end{matrix}$$

$$= \int_{-1}^1 \int_0^1 \frac{x^2 y}{\sqrt{1-x^2}} \, dy \, dx \quad \begin{matrix} x=0, \theta=0 \\ x=1, \theta=\pi/2 \end{matrix}$$

$$= \int_{-1}^1 \frac{x^2 y^2}{2\sqrt{1-x^2}} \, dx = 2 \int_0^1 \frac{x^2}{2\sqrt{1-x^2}} \, dx$$

$$= \int_0^{\pi/2} \frac{\sin^2 \theta \cdot \cos \theta}{\cos \theta} d\theta$$

$$= \int_0^{\pi/2} \sin^2 \theta d\theta = \frac{1}{2} \cdot \frac{\pi}{2} = \underline{\underline{\frac{\pi}{4}}}$$

10.

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Divergence Theorem (Gauss Theorem)

Flux: Rate of change of fluid/heat/electricity passing through unit area per unit time.

Let G be a solid whose surface σ is oriented outward. If $F(x, y, z) = f\hat{i} + g\hat{j} + h\hat{k}$ where f, g, h have continuous first partial derivatives on some open set containing G and if n is the outward unit normal on σ then

$$\text{Flux} = \iint_{\sigma} F \cdot n \, ds = \iiint_G \text{div } F \, dV$$

1. Use divergence theorem to find the outward flux of the vector field $F(x, y, z) = z\hat{k}$ across the sphere $x^2 + y^2 + z^2 = a^2$.

Ans By divergence theorem

2 Marks

$$\iint_{\sigma} F \cdot n \, ds = \iiint_G \operatorname{div} F \, dv$$

$$\operatorname{div} F = \nabla \cdot F$$

$$= \left(\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k \right) \cdot (0i + 0j + zk)$$

$$= 1$$

$$= \iiint 1 \, dv$$

= Volume of the sphere of radius a

$$= \frac{4}{3} \pi a^3$$

2. $F(x, y, z) = (x-z)i + (y-x)j + (2z-y)k$, σ is the surface of the cylindrical solid bdd by $x^2 + y^2 = a^2$, $z=0$, $z=1$

2 Marks

$$\begin{aligned} \text{Ans: } \iint F \cdot n \, ds &= \iiint \operatorname{div} F \, dv \\ &= \iiint (1+1+2) \, dv \end{aligned}$$

$$= 4 \iiint dV$$

= 4 Volume of the cylinder

= 4 (base area $\times$ height)

$$= 4 \pi a^2 \times 1$$

$$= \underline{\underline{4\pi a^2}}$$

3. Verify divergence theorem by evaluating the surface integral and the triple integral. $F(x, y, z) = xi + yj + zk$, σ is the surface of the cube bounded by the planes $x=0, x=1, y=0, y=1, z=0, z=1$

Ans: By ~~surface integral~~ divergence theorem.

$$\iint F \cdot nds = \iiint \text{div } F dV$$

$$= \iiint (1+1+1) dV$$

$$= 3 \iiint dV$$

= 3 Volume of the cube

$$= 3 \times 1 \times 1 \times 1$$

$$= \underline{\underline{3}}$$

4. Use divergence theorem to evaluate
~~Q~~ $\iint_{\sigma} F \cdot n \, ds$ where $F(x, y, z) = x^3 i + y^3 j + z^3 k$
 σ is the surface of the cylindrical
 solid bounded by $x^2 + y^2 = 4$, $z = 0$ & $z = 4$

Ans $\iint_{\sigma} F \cdot n \, ds = \iiint \operatorname{div} F \, dv$

$$= \iiint 3x^2 + 3y^2 + 3z^2 \, dz$$

$$= 3 \int_0^4 \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (x^2 + y^2 + z^2) \, dy \, dx \, dz$$

$$= 6 \int_0^4 \int_{-2}^2 \left[(x^2 + z^2)y + \frac{y^3}{3} \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \, dx \, dz$$

$$= 12 \int_0^4 \int_{-2}^2 (x^2 + z^2) \sqrt{4-x^2} + \frac{(\sqrt{4-x^2})^3}{3} \, dx \, dz$$

$$= 12 \int_0^4 \int_0^2 (x^2 + z^2) \sqrt{4-x^2} + \frac{(\sqrt{4-x^2})^3}{3} \, dx \, dz$$

put $x = 2 \sin \theta$
 $dx = 2 \cos \theta \, d\theta$

$x=0, \theta=0$
 $x=2, \theta=\pi/2$

$$= 12 \int_0^4 \int_0^{\pi/2} \left[(4 \sin^2 \theta + z^2) 2 \cos \theta + \frac{(2 \cos \theta)^3}{3} \right] 2 \cos \theta \, d\theta \, dz$$

$$= 12 \int_0^4 \int_0^{\pi/2} \left(4 \sin^2 \theta \cdot 2 \cos \theta + z^2 2 \cos \theta + \frac{16 \cos^3 \theta}{3} \right) d\theta \, dz$$

$$= 12 \int_0^4 \left(\frac{16 \cdot 1 \cdot 1 \cdot \pi}{4 \cdot 2 \cdot 2} + z^2 \cdot 4 \cdot \frac{1}{2} \cdot \frac{\pi}{2} + \frac{16 \cdot \frac{4}{3} \cdot 1 \cdot \pi}{3 \cdot 4 \cdot 2 \cdot 2} \right) dz$$

$$= 12 \int_0^4 \left(\pi + z^2 \pi + \frac{2}{3} \pi \right) dz$$

$$= 12 \left(\frac{2}{3} \pi z + \frac{\pi z^3}{3} \right) \Big|_0^4$$

$$= 12 \left(\frac{8}{3} + \frac{64}{3} \right) \pi$$

$$= \underline{\underline{352\pi}}$$

5. Use divergence theorem to find the outward flux of the vector field $F(x, y, z) = x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}$ across the surface of the region that is enclosed by the circular cylinder $x^2 + y^2 = 9$ & the plane $z=0$ & $z=2$

Ans: 315\pi

use divergence theorem to find the flux of
 6. $F(x, y, z) = (x^2 + y)\hat{i} + xy\hat{j} - (2xz + y)\hat{k}$
 across the surface of the tetrahedron
 in the first octant bounded by
 $x + y + z = 2$ and the co-ordinate planes.

Ans $\text{div } F = 2x + x - 2x = x$

$$\iint F \cdot n \, ds = \iiint x \, dv \quad \begin{array}{l} z: 0 \text{ to } 2-x-y \\ x: 0 \text{ to } 2 \\ y: 0 \text{ to } 2-x \end{array}$$

$$= \int_0^2 \int_0^{2-x} \int_0^{2-x-y} x \, dz \, dy \, dx$$

$$= \int_0^2 \int_0^{2-x} \left(xz \right)_{z=0}^{z=2-x-y} dy \, dx$$

$$= \int_0^2 \int_0^{2-x} x(2-x-y) dy \, dx$$

$$= \int_0^2 \left(x(2-x)y - \frac{xy^2}{2} \right)_{y=0}^{y=2-x} dx$$

$$= \int_0^2 x(2-x)^2 - \frac{x}{2}(2-x)^2 dx$$

$$= \int_0^2 \frac{x}{2}(4-4x+x^2) dx$$

$$= \int_0^2 2x - 2x^2 + \frac{x^3}{2} dx$$

$$= \left[\frac{2x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{8} \right]_0^2$$

$$= 2^2 - \frac{2^4}{3} + \frac{2^4}{8}$$

$$= 4 - \frac{16}{3} + 2 = \frac{12}{3} - \frac{16}{3} + \frac{6}{3} = \frac{18-16}{3}$$

$$= \frac{2}{3}$$

7. Evaluate $\iint_S F \cdot n \, ds$, where

$F(x, y, z) = 2x\hat{i} - yz\hat{j} + z^2\hat{k}$ the surface σ is the paraboloid $z = x^2 + y^2$ capped by the disc $x^2 + y^2 \leq 1$ in the plane $z = 1$

Ans $\iint_S F \cdot n \, ds = \iiint_V \text{div } F \, dV$

$$\text{div } F = 2 - z + 2z = z + 2$$

$$= \iiint_V (z + 2) \, dV$$

$$z: 1 \text{ to } x^2 + y^2$$

$$x: -1 \text{ to } 1$$

$$y: -\sqrt{1-x^2} \text{ to } \sqrt{1-x^2}$$

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_1^{x^2+y^2} (z+2) dz dy dx$$

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \left(\frac{z^2}{2} + 2z \right) \Big|_{z=1}^{z=x^2+y^2} dy dx$$

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \left(\frac{(x^2+y^2)^2}{2} + 2(x^2+y^2) - \frac{1}{2} - 2 \right) dy dx$$

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \left(\frac{(x^2+y^2)^2}{2} + 2(x^2+y^2) - \frac{5}{2} \right) dy dx$$

$$= 2 \int_{-1}^1 \left[\int_0^{\sqrt{1-x^2}} \left(\frac{x^4 + 2x^2y^2 + y^4}{2} + 2x^2 + 2y^2 - \frac{5}{2} \right) dy \right] dx$$

$$= 2 \int_{-1}^1 \left(\frac{x^4 + 2x^2 - 5}{2} y + \frac{x^2 y^3}{3} + \frac{y^5}{5} + 2y^3 \right) \Big|_0^{\sqrt{1-x^2}} dx$$

$$= 2 \int_{-1}^1 \left(\frac{x^4 + 2x^2 - 5}{2} \sqrt{1-x^2} + \frac{x^2 (\sqrt{1-x^2})^3}{3} + \frac{(\sqrt{1-x^2})^5}{5} + \frac{2}{3} (\sqrt{1-x^2})^3 \right) dx$$

$$= 4 \int_0^1 \left(\frac{x^4 \sqrt{1-x^2}}{2} + \frac{1}{3} x^3 (\sqrt{1-x^2})' + \frac{1}{10} (\sqrt{1-x^2})^5 + 2x^2 \sqrt{1-x^2} + \frac{2}{3} (\sqrt{1-x^2})^3 - \frac{5}{2} \sqrt{1-x^2} \right) dx$$

$$= 4 \int_0^{\pi/2} \frac{\sin^4 \theta \cos^2 \theta}{2} + \frac{\sin^2 \theta \cos^4 \theta}{3} + \frac{\cos^6 \theta}{10} + \left[\begin{array}{l} x = \sin \theta \\ dx = \cos \theta d\theta \\ \sqrt{1-x^2} = \cos \theta \end{array} \right]$$

$$2 \sin^2 \theta \cdot \cos^2 \theta + \frac{2}{3} \cos^4 \theta - \frac{2}{5} \cos^2 \theta$$

$$= 4 \left[\frac{1}{2} \left(\frac{3 \cdot 1 \cdot 1}{6 \cdot 4 \cdot 2} \right) + \frac{1}{3} \left(\frac{1 \cdot 3}{6 \cdot 4 \cdot 2} \right) + \frac{1}{10} \left(\frac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2} \right) + 2 \left(\frac{1 \cdot 1}{4 \cdot 2} \right) + \frac{2}{3} \left(\frac{3 \cdot 1}{4 \cdot 2} \right) - \frac{5}{2} \left(\frac{1}{2} \right) \right] \frac{\pi}{2}$$

$$= 2\pi \left[\frac{2}{32} + \frac{1}{48} - \frac{3}{4} \right]$$

$$= \underline{\underline{-\frac{4}{3}\pi}}$$

11Q

8) If σ is any closed surface enclosing a volume V , and $F = 2x\hat{i} + 2y\hat{j} + 3z\hat{k}$ using divergence theorem show that $\iint_{\sigma} F \cdot n ds = 7V$

$$\iint_{\sigma} F \cdot n ds = \iiint_V \text{div } F dV$$

$$= \iiint_G 2+2+3 \, dv$$

$$= 7 \iiint_G dv = \underline{\underline{7V}}$$

- 9) Use divergence theorem to find the flux F across the surface σ with outward orientation
 $F(x, y, z) = (x^2 + y) \mathbf{i} + z^2 \mathbf{j} + (e^y - z) \mathbf{k}$
 σ is the surface of the rectangular solid bounded by the co-ordinate planes & the planes $x=3, y=1, z=3$

$$\iint_{\sigma} F \cdot n \, ds = \iiint_G \operatorname{div} F \, dv$$

$$= \iiint_G (2x-1) \, dv$$

$$= \int_0^3 \int_0^1 \int_0^3 (2x-1) \, dx \, dy \, dz$$

$$= \underline{\underline{18}}$$

STOKES THEOREM

It is the generalisation of Green's Theorem.

Let σ be a piecewise smooth oriented surface that is bounded by a simple closed piecewise smooth curve C with +ve orientation. If the components of the vector field $F(x, y, z) = f\hat{i} + g\hat{j} + h\hat{k}$ are continuous and have continuous first order partial derivatives on some open set containing σ , then

Work done =
$$\oint_C F \cdot d\mathbf{r} = \oint_C F \cdot T ds = \iint_{\sigma} \text{Curl } F \cdot \mathbf{n} ds$$

where $\mathbf{n}$ is the unit ^{normal} vector at any point of S drawn in the sense in which a right handed screw ~~will~~ would advance when rotated in the sense of description of C .

Note To calculate $n \, ds$.

Case ① +vely oriented surface
(oriented outwards)
(counter clockwise direction)
+ve z direction

Take the surface as

$$G = z - g(x, y)$$

$$\text{find } \nabla G = \frac{\partial G}{\partial x} \hat{i} + \frac{\partial G}{\partial y} \hat{j} + \frac{\partial G}{\partial z} \hat{k}$$

$$\text{unit normal vector } \hat{n} = \frac{\nabla G}{\|\nabla G\|}$$

$$ds = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}$$
$$= \|\nabla G\|$$

$$n \, ds = \frac{\nabla G}{\|\nabla G\|} \|\nabla G\| \, dA$$

$$= \nabla G \, dA$$

$$\therefore \text{Work done} = \iint \text{Curl } F \cdot \nabla G \, dA$$

Case ② -vely oriented surface
(oriented inwards)
(clockwise direction)
(-ve z direction)

Take the surface as

$$G = g(x, y) - z$$

$$n = \frac{\nabla G}{\|\nabla G\|}$$

$$nds = \frac{\nabla G}{\|\nabla G\|} \|\nabla G\| dA$$

$$= \nabla G dA$$

$$\therefore \boxed{\text{Work done} = \iint \text{Curl } F \cdot \nabla G dA}$$

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1) Find the work done performed by the force field $F(x, y, z) = 3z\hat{i} + 4x\hat{j} + 2y\hat{k}$ C is the boundary of the paraboloid $z = 4 - x^2 - y^2$ oriented outwards.

Ans: By Stokes' Theorem

$$\text{Work done} = \oint_C F \cdot d\mathbf{r} = \iint_S \text{Curl } F \cdot \nabla G \, dA$$

$$\text{Curl } F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3z & 4x & 2y \end{vmatrix}$$

$$= \hat{i}(2-0) - \hat{j}(0-3) + \hat{k}(4-0)$$

$$= 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\text{Let } G = z - 4 + x^2 + y^2 \quad (z \text{ should be +ve because oriented outwards})$$

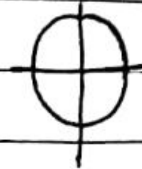
$$\nabla G = 2x\hat{i} + 2y\hat{j} + \hat{k}$$

$$W = \iint (2\hat{i} + 3\hat{j} + 4\hat{k}) \cdot (2x\hat{i} + 2y\hat{j} + \hat{k}) dA$$

$$= \iint 4x + 6y + 4 dA$$

$$= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4x + 6y + 4) dy dx$$

$x^2 + y^2 = 4$



$$= \int_{-2}^2 2 \int_0^{\sqrt{4-x^2}} 4(x+1) dy dx$$

$$= \int_{-2}^2 8(x+1)\sqrt{4-x^2} dx$$

$$= 8 \int_{-2}^2 \left(\underset{\text{odd}}{x\sqrt{4-x^2}} + \underset{\text{even}}{\sqrt{4-x^2}} \right) dx$$

$$= 16 \int_0^2 \sqrt{4-x^2} dx$$

$$= 16 \cdot \left(\frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right) \Big|_0^2$$

$$= 16 (0 + 2 \sin^{-1} 1)$$

$$= 16 \times 2 \cdot \frac{\pi}{2} = \underline{\underline{16\pi}}$$

2. Find the work done performed by the force field $F(x, y, z) = (x-2y)\hat{i} + (y-z)\hat{j} + (z-x)\hat{k}$, where C is the circle $x^2 + y^2 = a^2$ in the xy plane with counterclockwise direction looking down the z axis.

Ans:

$$\oint_C F \cdot d\vec{s} = \iint \text{curl } F \cdot \vec{n} \, ds$$

eqn of the surface
 $G = z = 0$
is xy plane

$$= \iint \text{curl } F \cdot \nabla G \, dA$$

~~the surface~~

$$\nabla G = \hat{k}$$

$$= \iint (\hat{i} + \hat{j} + 2\hat{k}) \cdot \hat{k} \, dA$$

curl F

$$= \iint 2 \, dA$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x-2y & y-z & z-x \end{vmatrix}$$

$$= 2 \iint dA$$

$$= 2 \cdot \text{Area of } x^2 + y^2 = a^2 \quad \& \quad \begin{pmatrix} i(0+1) - j(-1+0) + k(2) \\ = \hat{i} + \hat{j} + 2\hat{k} \end{pmatrix}$$

$$= 2 \cdot \pi a^2$$

$$= \underline{\underline{2\pi a^2}}$$

3 Use Stokes' Theorem to evaluate $\oint_C F \cdot d\mathbf{r}$
 $F(x, y, z) = z^2 \mathbf{i} + 3xz \mathbf{j} - y^3 \mathbf{k}$; C is the
 circle $x^2 + y^2 = 1$ in the xy plane
 with counterclockwise orientation
 looking down the ⁺ z axis

Ans:- $\oint_C F \cdot d\mathbf{r} = \iint \text{Curl } F \cdot \mathbf{n} \, ds$

$$\begin{aligned} \text{Curl } F &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & 3xz & -y^3 \end{vmatrix} \\ &= \hat{i}(-3y^2) - \hat{j}(-2z) + \hat{k}(3) \\ &= -3y^2 \hat{i} + 2z \hat{j} + 3 \hat{k} \end{aligned}$$

Let $G = xy$ plane $G = z = 0$
 $\nabla G = \hat{k}$

$$\begin{aligned} \oint_C F \cdot d\mathbf{r} &= \iint (-3y^2 \hat{i} + 2z \hat{j} + 3 \hat{k}) \cdot \hat{k} \, dA \\ &= 3 \iint dA \\ &= 3 \text{ Area of } x^2 + y^2 = 1 \\ &= \underline{\underline{3\pi}} \end{aligned}$$

4. Use Stokes theorem to evaluate $\oint_C F \cdot d\mathbf{r}$, where $F(x, y, z) = -3y^2\mathbf{i} + 4z\mathbf{j} + 6x\mathbf{k}$, where C is the triangle in the plane $z = \frac{y}{2}$ with vertices $(2, 0, 0)$, $(0, 2, 1)$ & $(0, 0, 0)$ with a counter-clockwise orientation looking down the +ve z axis.

$$\text{Ans: } W = \oint_C F \cdot d\mathbf{r} = \iint_S \text{Curl } F \cdot \mathbf{n} \, dA$$

$$= \iint_S \text{Curl } F \cdot \nabla G \, dA$$

$$= \iint_S (-4\mathbf{i} - 6\mathbf{j} + 6y\mathbf{k}) \cdot \left(-\frac{1}{2}\mathbf{j} + \mathbf{k}\right) dA$$

$\text{Curl } F =$	$\mathbf{i}$	$\mathbf{j}$	$\mathbf{k}$
	$\frac{\partial}{\partial x}$	$\frac{\partial}{\partial y}$	$\frac{\partial}{\partial z}$
	$-3y^2$	$4z$	$6x$

$$= \iint_S 3 + 6y \, dA$$

$$= \int_0^2 \int_0^{2-x} 3 + 6y \, dy \, dx$$

$$= \int_0^2 \left(3y + 6y^2\right) \Big|_0^{2-x} dx$$

$$= \int_0^2 3(2-x) + 3(2-x)^2 \, dx$$

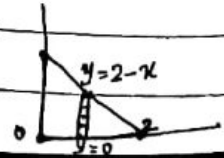
$$= \left[\frac{3(2-x)^2}{-2} + \frac{3(2-x)^3}{-3} \right]_0^2 = 6 + 8 = 14$$

$$= \mathbf{i}(-4) - \mathbf{j}(6) + \mathbf{k}(+6y)$$

$$= -4\mathbf{i} - 6\mathbf{j} + 6y\mathbf{k}$$

$$\text{Let } G = z - y/2$$

$$\nabla G = -\frac{1}{2}\mathbf{j} + \mathbf{k}$$



5. Verify Stokes' theorem by evaluating the line integral and the surface integral. Assume that the surface has an upward orientation. $F(x, y, z) = (x-y)\hat{i} + (y-z)\hat{j} + (z-x)\hat{k}$, σ is the portion of the plane $x+y+z=1$ in the first octant.

$$W = \oint F \cdot d\mathbf{r} = \iint \text{Curl } F \cdot \mathbf{n} \, d\mathbf{s}$$

Verification
is difficult

~~$$\oint F \cdot d\mathbf{r} = \oint f \, dx + g \, dy + h \, dz$$~~

Ans $\text{RHS} = \iint \text{Curl } F \cdot \nabla G \, dA$ $G = z + x + y - 1$
 $= \iint (i+j+k) \cdot (i+j+k) \, dA$ $\nabla G = i+j+k$

$$= \iint 1+1+1 \, dA$$

$$\text{Curl } F = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x-y & y-z & z-x \end{vmatrix}$$

$$= \iint 3 \, dA$$

$$= i(0+1) - j(-1) + k(1)$$

$$= i + j + k$$

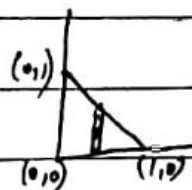
$$= 3 \iint dA$$

$$= 3 \times \text{Area of the } \Delta \text{ in the } xy \text{ plane}$$

$$= 3 \cdot \frac{1}{2}bh$$

$$= 3 \times \frac{1}{2} \times 1 \times 1$$

$$= \underline{\underline{\frac{3}{2}}}$$



Q12

6. Apply Stokes theorem to evaluate $\oint_C (x+y)dx + (2x-y)dy + (y+z)dz$ where C is the bdy of the triangle with vertices $(0,0,0)$, $(2,0,0)$, $(0,3,0)$

Ans- $W = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dA$

$$= \iint_S \text{curl } \mathbf{F} \cdot \nabla G dA$$

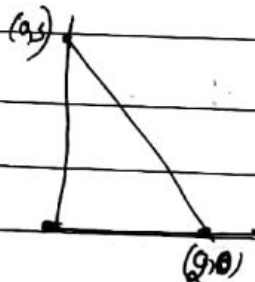
$$= \iint_S (\mathbf{i} + \mathbf{k}) \cdot \mathbf{k} dA$$

$$= \iint_S dA$$

Area of the Δ in the XY plane is always zero because 3rd component is always zero

$$= \frac{1}{2} \times 2 \times 3$$

$$= 3$$



$$\nabla G = \mathbf{k}$$

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y & 2x-y & y+z \end{vmatrix}$$

$$= \mathbf{i}(1) - \mathbf{j}(0) + \mathbf{k}(2-1)$$

$$= \mathbf{i} + \mathbf{k}$$

Sources & Sinks of a vector field

The points at which $\text{div } F \geq 0$ are called sources and points at which $\text{div } F < 0$ are called sinks.
If $\text{div } F = 0$, the fluid has no sources & no sinks.

- 1) Determine whether the vector field $F(x, y, z)$ is free of sources & sinks. If it is not locate them.
 $F(x, y, z) = (y+z)\hat{i} + xz^3\hat{j} + x^2\sin y\hat{k}$

Ans. - $\text{div } F = 0\hat{i} + 0\hat{j} + 0\hat{k} = 0$

$\text{div } F = 0$ no sources & sinks

2) $F(x, y, z) = x^3\hat{i} + y^3\hat{j} + 2z^3\hat{k}$

$$\text{div } F = 3x^2\hat{i} + 3y^2\hat{j} + 6z^2\hat{k} > 0$$

for all except at origin.

$\therefore$ We have sources at all points except at origin. $\text{div } F$ cannot be -ve and hence has no sinks.

3) $F(x, y, z) = xy \hat{i} - 2xy \hat{j} + y^2 \hat{k}$
 Ans:- $\text{div } F = y - 2x + 0$

$\text{div } F > 0 \rightarrow y - 2x > 0$

$\Rightarrow y > 2x$

$\text{div } F < 0 \Rightarrow y - 2x < 0$

$y < 2x$

$\therefore$ Sources where $y > 2x$ & sinks where $y < 2x$.

4) $F(x, y, z) = 2(x^3 - 2x) \hat{i} + 2(y^3 - 2y) \hat{j} + 2(z^3 - 2z) \hat{k}$

$\text{div } F = 2(3x^2 - 2) + 2(3y^2 - 2) + 2(3z^2 - 2)$

$= 6(x^2 + y^2 + z^2) - 12$

$= x^2 + y^2 + z^2 - 2$

$\text{div } F > 0 \Rightarrow x^2 + y^2 + z^2 > 2$

i.e sources outside the sphere $x^2 + y^2 + z^2 = 2$

$\text{div } F < 0 \Rightarrow x^2 + y^2 + z^2 < 2$

i.e sinks, inside the sphere $x^2 + y^2 + z^2 = 2$.

Reduction formula

Case 1

$$\int_0^{\pi/2} \cos^n \theta d\theta \quad \text{OR} \quad \int_0^{\pi/2} \sin^n \theta d\theta$$

$$\left. \begin{array}{l} \text{Formula } \int_0^{\pi/2} \cos^n \theta d\theta \\ \int_0^{\pi/2} \sin^n \theta d\theta \end{array} \right\} = \frac{(n-1)(n-3)(n-5)\dots 1 \text{ OR } 2}{n(n-2)(n-4)\dots 1 \text{ OR } 2} \times k$$

$$\text{where } k = \begin{cases} \frac{\pi}{2} & \text{if } n \text{ is even} \\ 1 & \text{if } n \text{ is odd} \end{cases}$$

eg: 1) $\int_0^{\pi/2} \sin^7 \theta d\theta = \int_0^{\pi/2} \cos^7 \theta d\theta = \frac{6 \times 4 \times 2}{7 \times 5 \times 3 \times 1} \left(\begin{array}{l} \text{no need to} \\ \text{multiply with } \pi/2 \\ \text{because 7 is odd} \end{array} \right)$

2) $\int_0^{\pi/2} \sin^{10} \theta d\theta = \int_0^{\pi/2} \cos^{10} \theta d\theta = \frac{9 \times 7 \times 5 \times 3 \times 1}{10 \times 8 \times 6 \times 4 \times 2} \times \frac{\pi}{2} \left(\begin{array}{l} \text{Since 10 is an} \\ \text{even number} \\ \text{multiply with } \pi/2 \end{array} \right)$

3) $\int_0^{\pi/2} \cos^5 \theta d\theta = \frac{4 \times 2 \times 0}{5 \times 3 \times 1}$

4) $\int_0^{\pi/2} \sin^8 \theta d\theta = \frac{7 \times 5 \times 3 \times 1}{8 \times 6 \times 4 \times 2} \times \frac{\pi}{2}$

Reduction formula for $\int_0^{\pi/2} \sin^n \theta \cdot \cos^m \theta d\theta$

Eg:- 1) $\int_0^{\pi/2} \sin^6 \theta \cdot \cos^7 \theta d\theta = \frac{(\quad)(\quad)}{(\quad)}$

$$= \frac{(5)(6)}{13 \times}$$

$$= \frac{(5 \times 3 \times 1)(6 \times 4 \times 2)}{13 \times 11 \times 9 \times 7 \times 5 \times 3 \times 1}$$

2) $\int_0^{\pi/2} \sin^8 \theta \cdot \cos^{10} \theta d\theta = \frac{(7 \times 5 \times 3 \times 1)(9 \times 7 \times 5 \times 3 \times 1)}{18 \times 16 \times 14 \times 12 \times 10 \times 8 \times 6 \times 4 \times 2} \times \frac{\pi}{2}$

3) $\int_0^{\pi/2} \cos^5 \theta \cdot \sin^3 \theta d\theta = \frac{(4 \times 2)(3 \times 1)}{8 \times 6 \times 4 \times 2}$

Module: 2 Summary.

I Green's Theorem (Work done in 2D)

$$W = \oint_C \mathbf{F} \cdot d\mathbf{x} = \oint_C f dx + g dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

C : closed curve
 R : Boundary of the closed region.

II Area using Green's Theorem.

$$\text{Area} = \frac{1}{2} \oint_C -y dx + x dy$$

III Divergence Theorem (Flux)

$$\text{Flux} = \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} d\sigma = \iiint_G \text{div } \mathbf{F} dV$$

$\mathbf{n}$: outward unit vector.

σ : Surface of the solid

G : Solid.

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$$

$$= \left(\frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \cdot (f \mathbf{i} + g \mathbf{j} + h \mathbf{k})$$

IV Stokes' Theorem (Work done in 3D) Generalisation of Green's Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{x} = W = \iint_{\sigma} \text{Curl } \mathbf{F} \cdot \nabla G dA$$

σ : smooth oriented surface

$\nabla G \Rightarrow$ gradient of G

$$\nabla G = \frac{\partial G}{\partial x} \mathbf{i} + \frac{\partial G}{\partial y} \mathbf{j} + \frac{\partial G}{\partial z} \mathbf{k}$$

$$\text{Curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix}$$

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